

CHARACTERIZATION OF CRYSTAL EFFICIENCY:

Methodology, Corrections, and Results

JULY 2026

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1. DATA ANALYZED

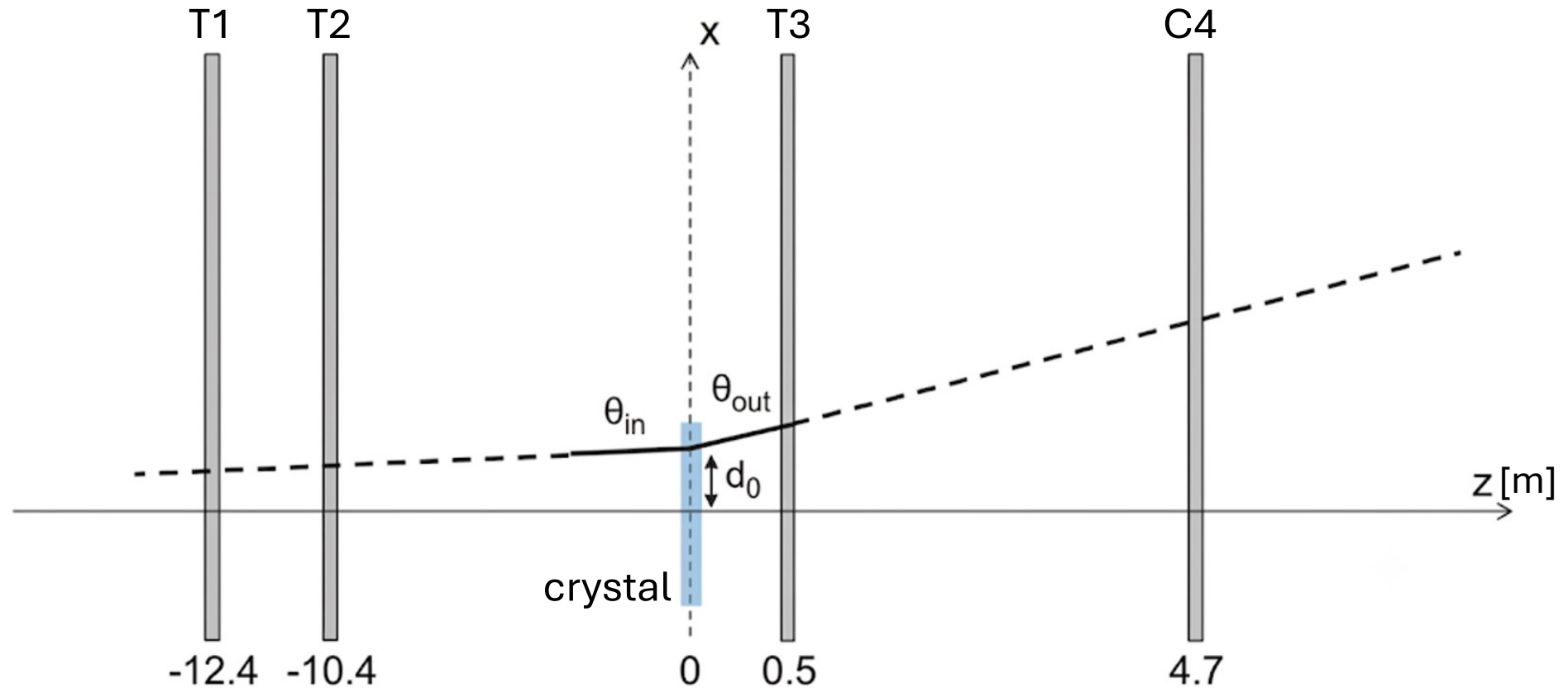
Analysis of 2024 dataset for two long crystals:

- LHCTBMCP0001-0001: runs 8430, 8431, 8650
- LHCTBMCP0001-0002: runs 8655, 8656

Configuration:

- Central goniometer
- Energy 180 GeV/c

1. SCHEMATIC LAYOUT



2. PREVIOUS ANALYSIS



SELECT SINGLE TRACKS

Selection of single tracks to eliminate multi-scattering events and multi-hit events (pile-up effect) from the dataset.



PARTICLE ENTRY

Validation of which particles successfully entered the crystal.

Since we do not know the relative beam position: algorithmic identification of the physical crystal edges based on interaction density profiles.



DETECTOR ALIGNMENT

Ensuring precision and the alignment of the detectors.

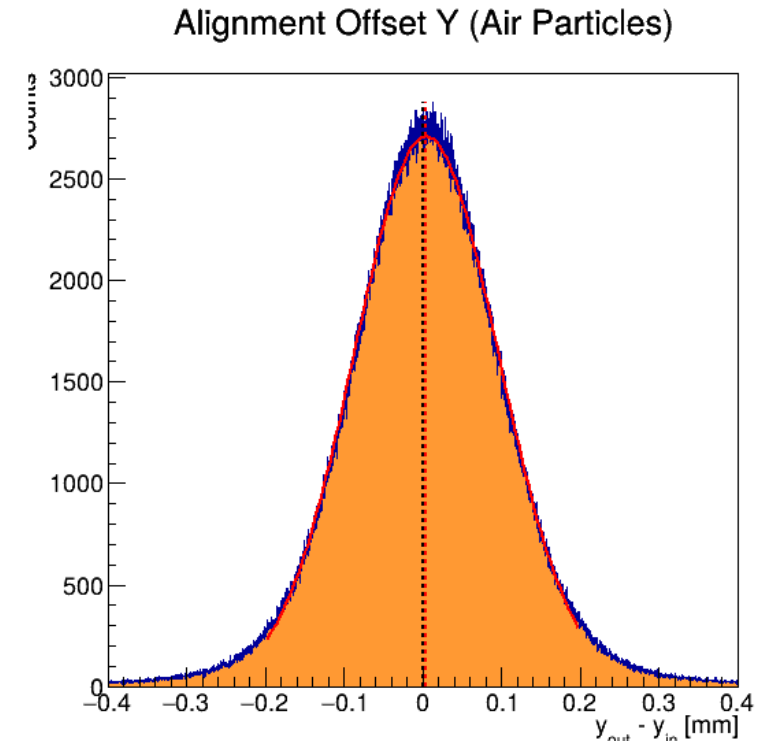
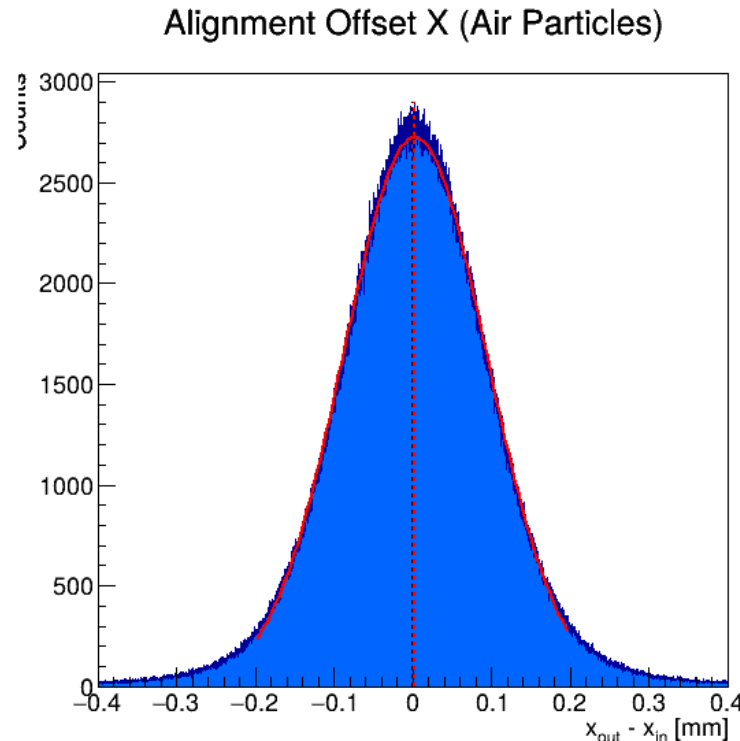
Any misalignment between tracking planes corrupts the angular measurements, necessitating corrections.

2. PREVIOUS ANALYSIS: ALIGNMENT

To ensure that the upstream and downstream detectors are aligned, we check the deviation between the initial and final positions of particles that definitely did not hit the crystal, ensuring that is zero.

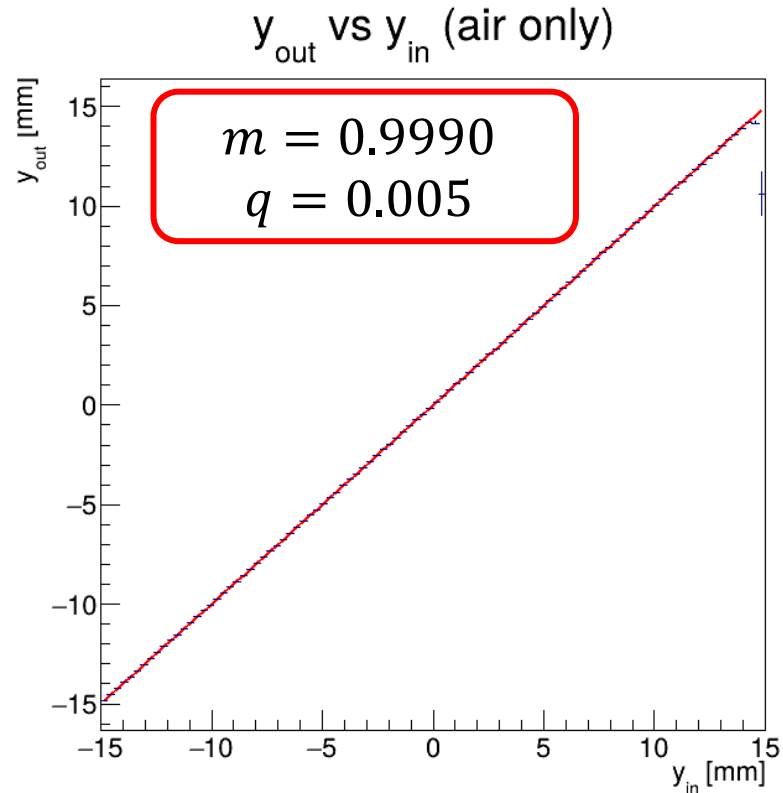
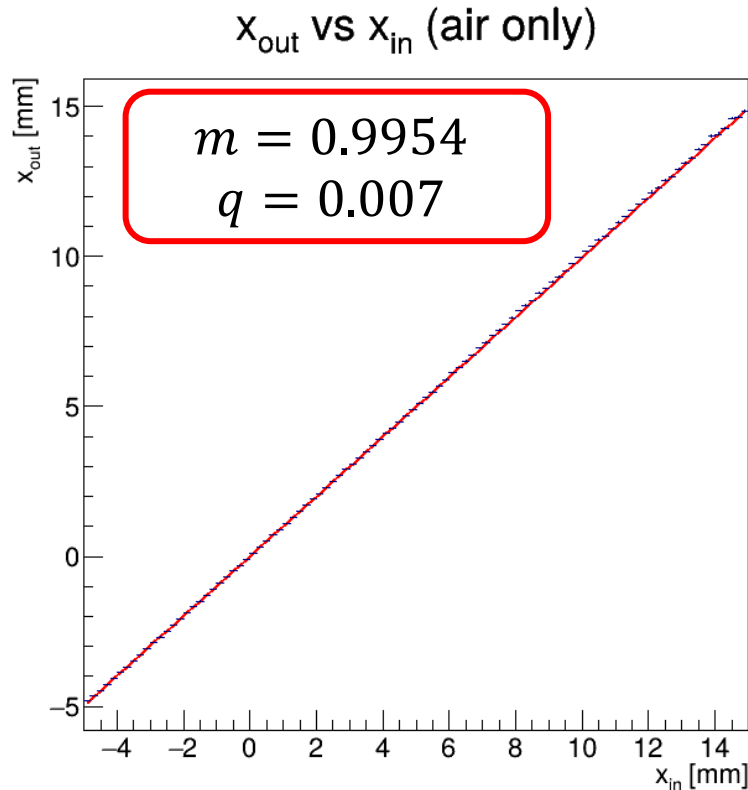
$\text{offset}_x = 0.003 \text{ mm}$
skewness ~ 0

$\text{offset}_y = 0.004 \text{ mm}$
skewness ~ 0



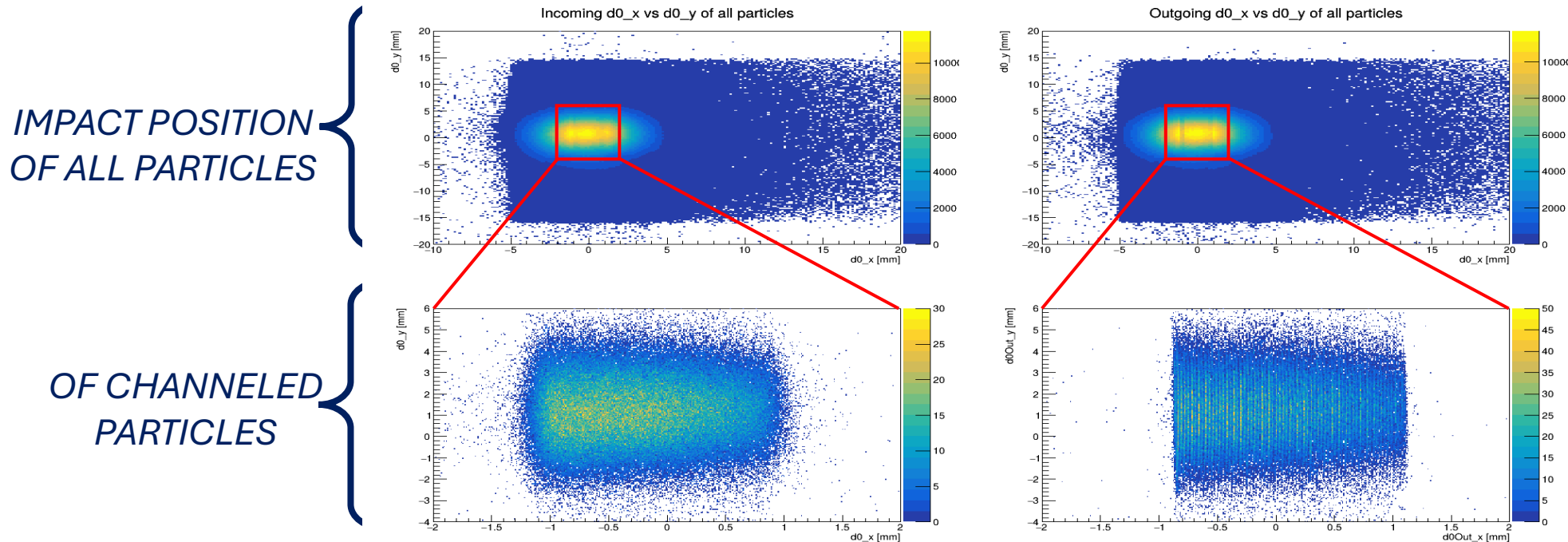
BACKUP SLIDES: ALIGNMENT

We can also check the alignment by doing a linear fit between the upstream and downstream profiles.



2. PREVIOUS ANALYSIS: CRYSTAL EDGES

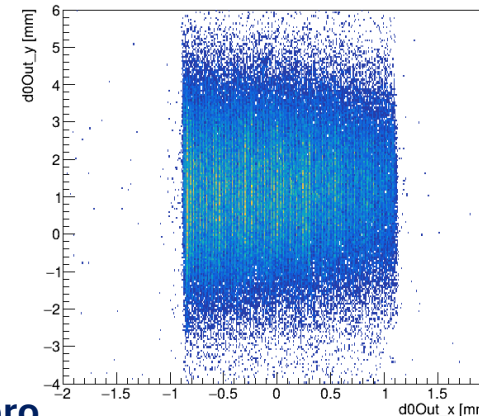
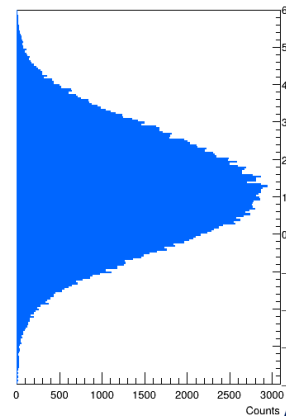
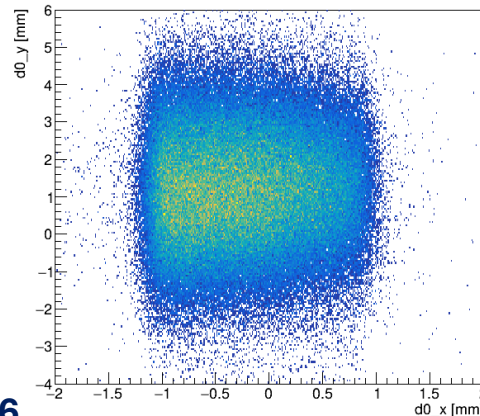
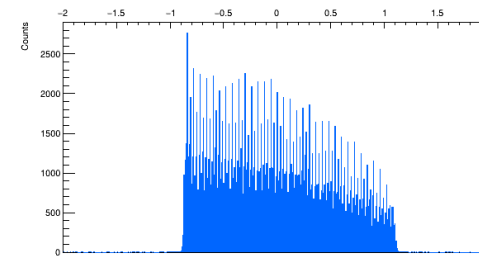
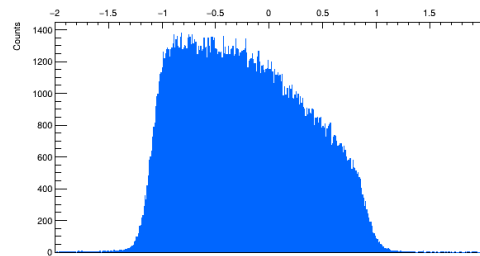
To select the particles that actually enter the crystal we first need to know the position of the crystal with respect to the beam.



We then look at channeled particles only using a preliminary deflection cut. Their impact points reveal the crystal's physical location, matching its expected dimensions.

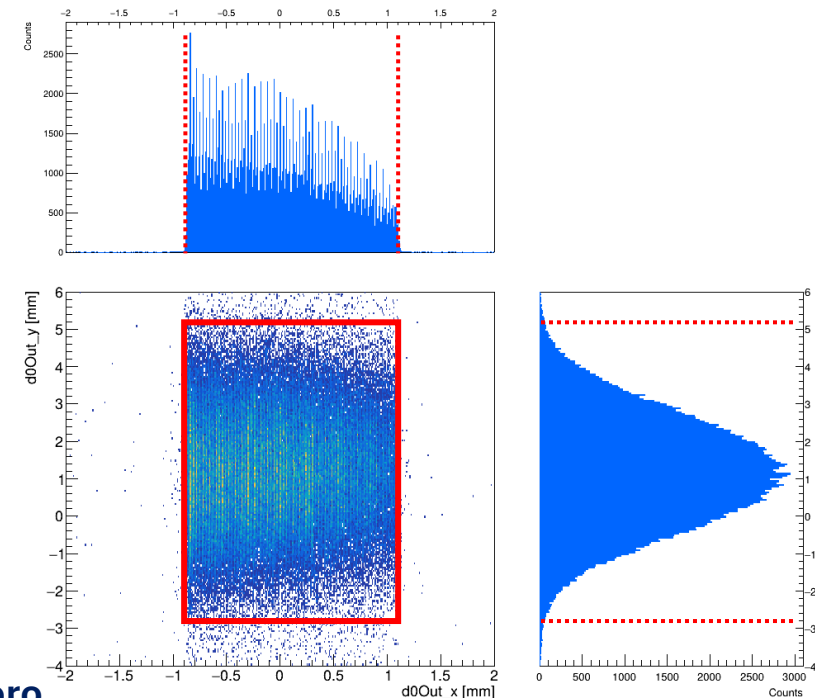
2. PREVIOUS ANALYSIS: CRYSTAL EDGES

- The impact points of these particles revealed the physical location of the crystal, matching its expected dimensions of 2 by 8 millimeters.
- To find the edges along x (that have an asymmetrical distribution) we perform a «sliding window» method (fixed width & precision of 0.01mm), aiming for the position that maximize the number of particles inside.



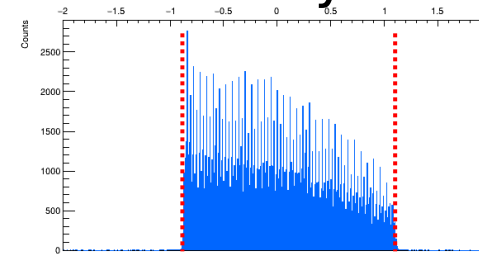
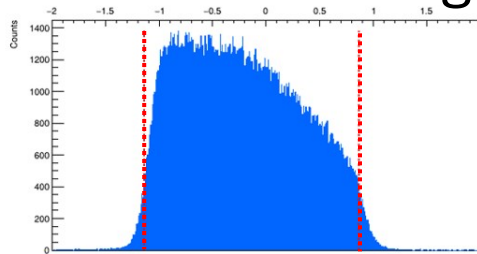
2. PREVIOUS ANALYSIS: CRYSTAL EDGES

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- To find the edges along x (that have an asymmetrical distribution) we perform a «sliding window» method (fixed width & precision of 0.01mm), aiming for the position that maximize the number of particles inside.
- The method was performed on the downstream particles, since they show more defined edges.



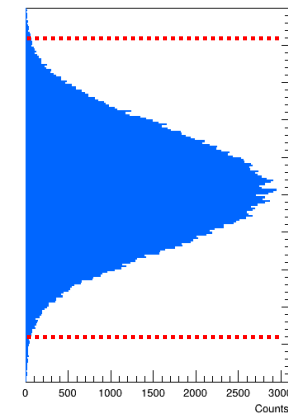
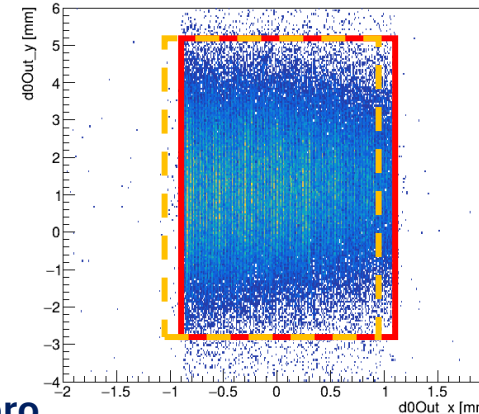
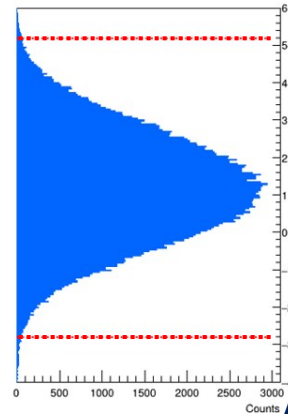
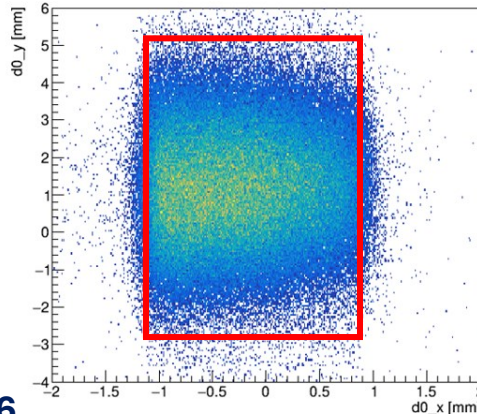
2. PREVIOUS ANALYSIS: CRYSTAL EDGES

- To define the true footprint at the entrance face we must project the downstream edges back upstream.
- We need to subtract a shift Δx from the downstream edge position.
- Channeled particles follow the bent lattice planes, resulting in a lateral geometric drift along the x-axis as they traverse the crystal.



$$\Delta x = \frac{\theta_b \cdot L}{2}$$

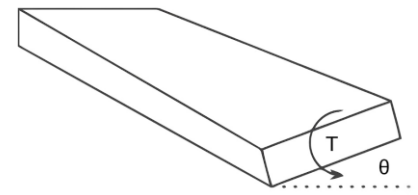
$\sim 0.22 \text{ mm}$



3. ANALYSIS: ANGLE CORRECTION

- The different orientations of the crystallographic planes along the crystal front surface influence the angular range in which particles can be channelled.
- Before selecting the particles that have entered with the right angle (within $\pm \frac{1}{2} \theta_L$) we need to apply a correction θ_0 because the cut may not be centered at zero.
- We do not choose a single θ_0 value but rather $\theta_0(x, y)$ in order to also apply a torsion correction which is not negligible in the long crystals.

3. ANALYSIS: TORSION EFFECT

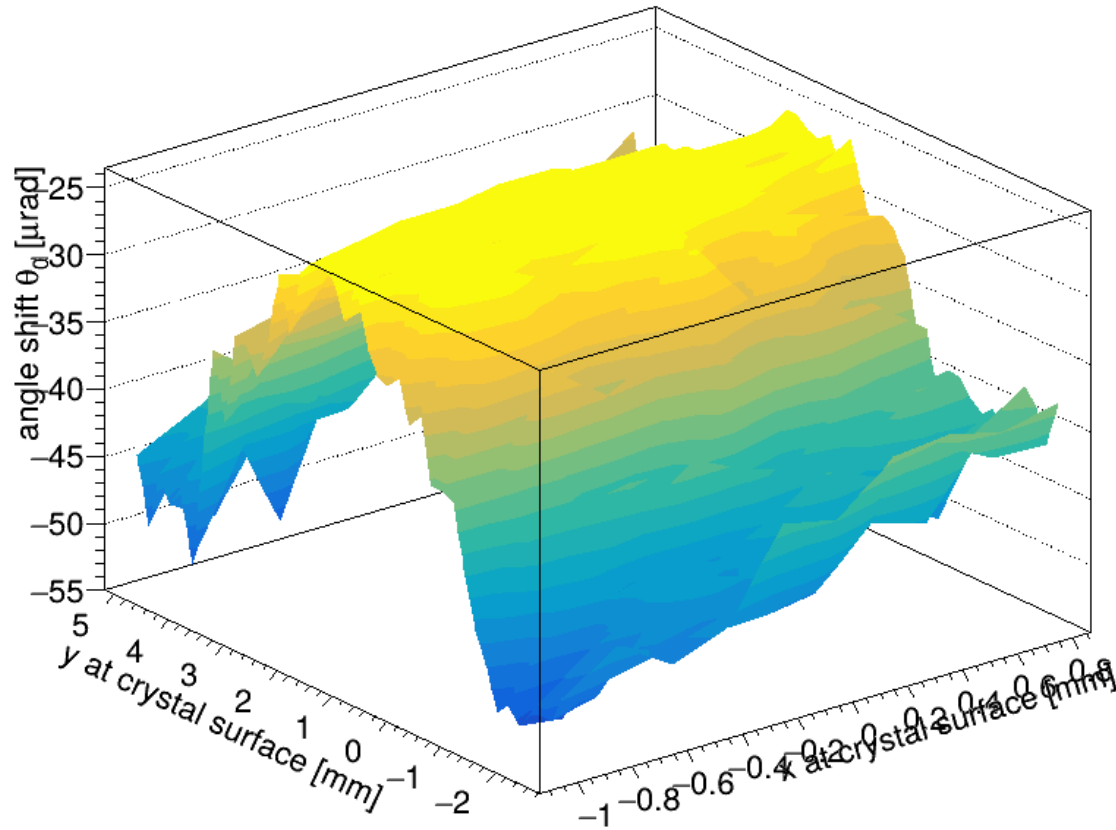


Torsion effect: in long crystals (>1 cm), structural torsion is non-negligible and must be incorporated into the analysis.

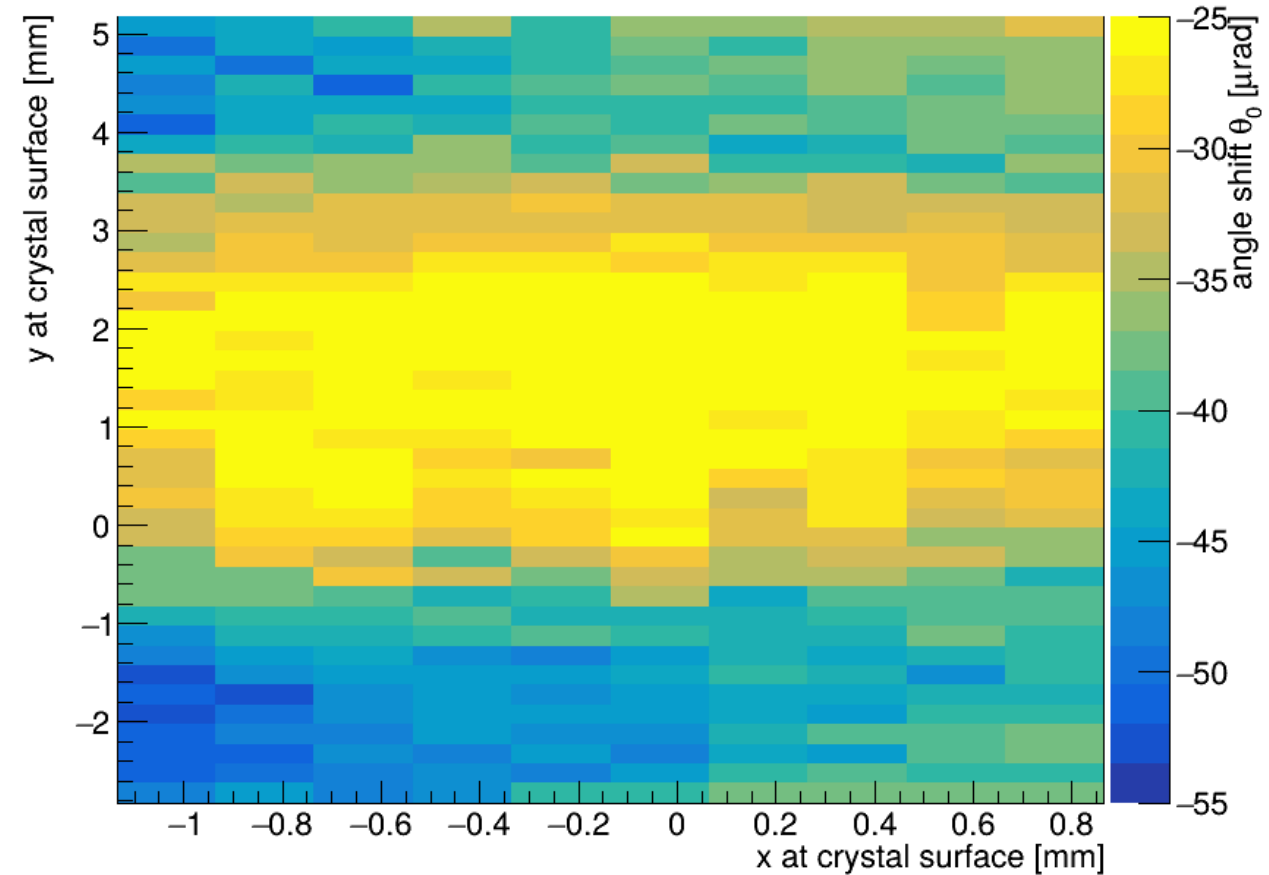
- The effect of the torsion can be corrected by doing a checkerboard analysis:
 1. Dividing the data sample into square bins (for example 0.1 mm x 0.1 mm) as a function of the impact position on the crystal surface
 2. The angular shift parameter θ_0 is then varied to find the maximum channeling efficiency ϵ_{ch} for each bin
 3. We obtain a 2D map of torsion variation across the crystal surface

3. ANALYSIS: TORSION EFFECT

2D Torsion Map (bins 0.1 x 0.1 mm)



2D Torsion Map (bins 0.1 x 0.1 mm)

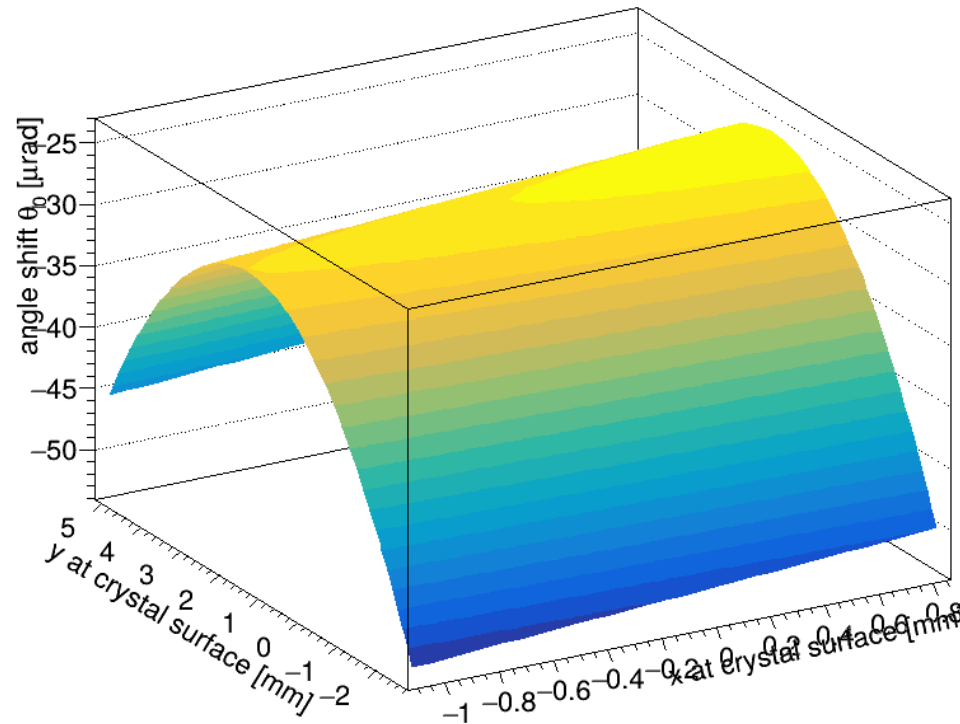


3. ANALYSIS: TORSION FIT

The 2d map is fitted through a parabolic function:

$$\theta_{corr}(x, y) = \theta_0 + \tau_x \cdot x + \tau_y \cdot y + p_3 y^2$$

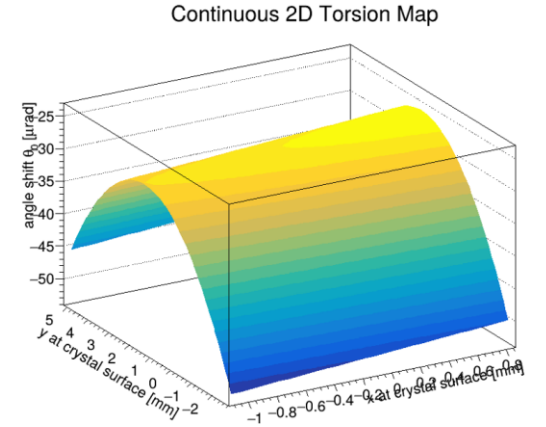
Continuous 2D Torsion Map



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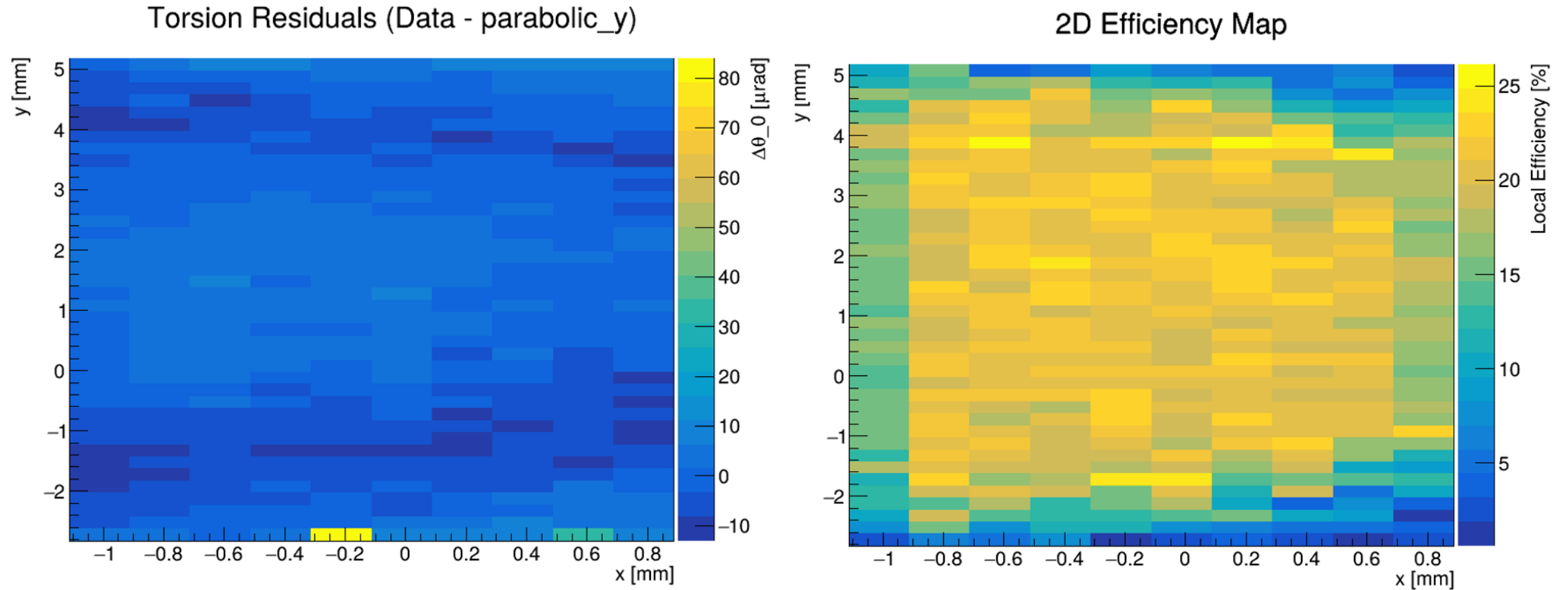


Now we correct the effect of torsion using the parameter $\theta_0(x, y)$ found for each bin.

instead of the Lindhard cut $|\theta_{in,x}| \leq \frac{1}{2} \theta_L$ we will use

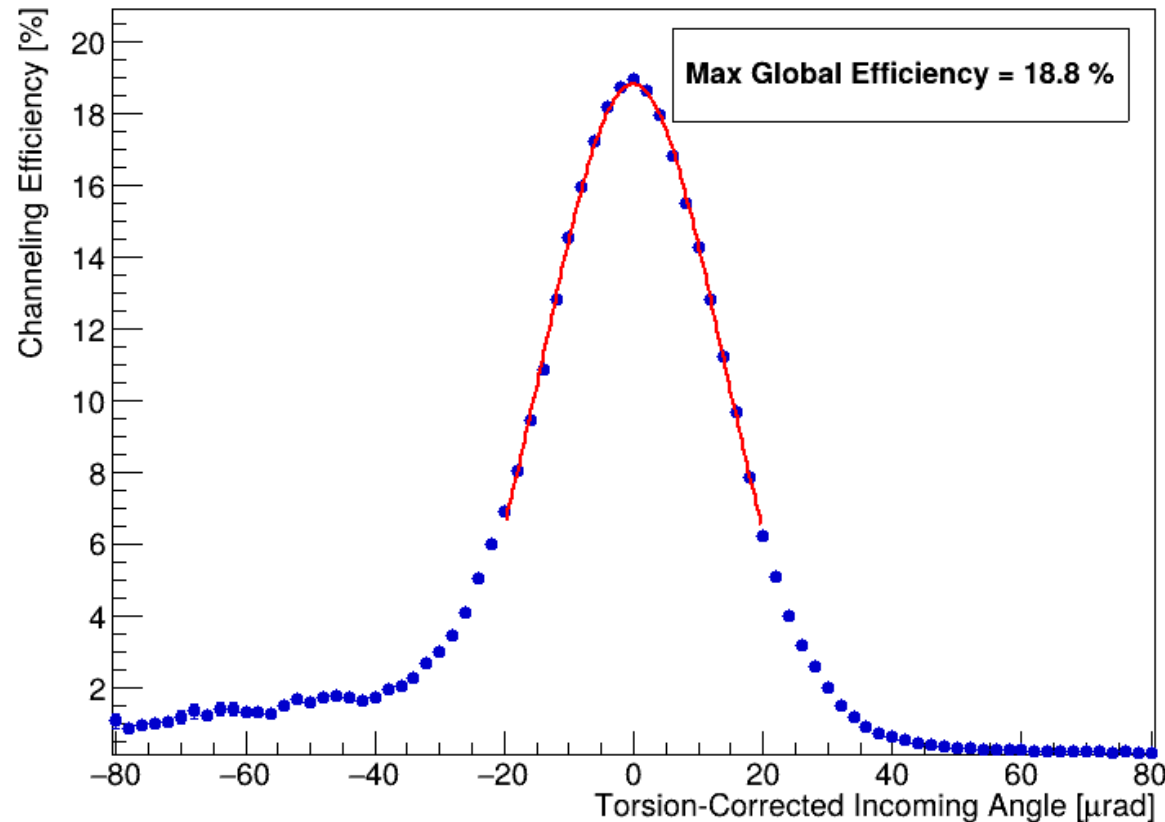
$$\left| \theta_x^{in} - \left(\theta_x^{off} + \tau_y \cdot d_y + \tau_x \cdot d_x \right) \right| \leq \frac{1}{2} \theta_L$$

3. ANALYSIS: TORSION FIT



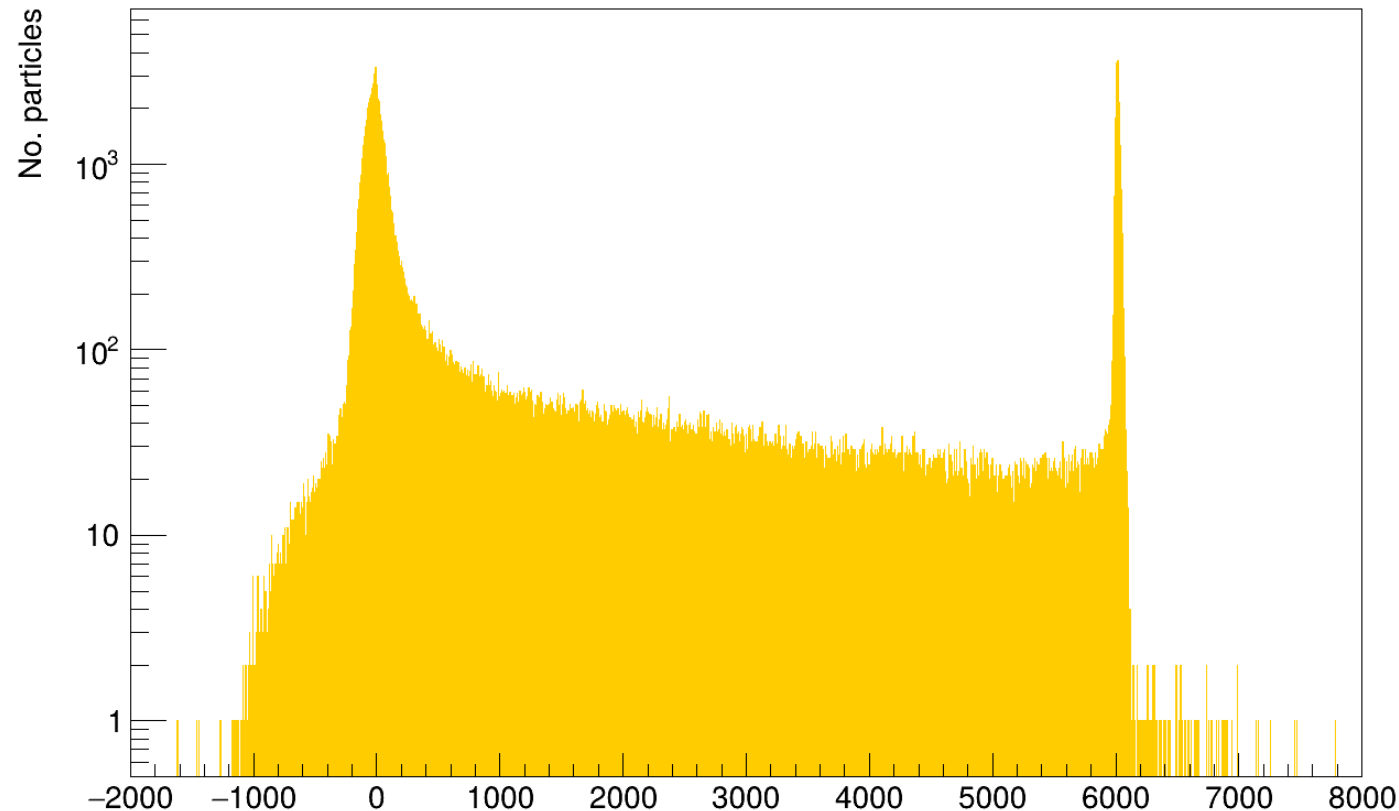
3. ANALYSIS: EFFICIENCY CURVE

Crystal efficiency as a function of the incoming angle, after application of the torsion correction. The value of the peak is the efficiency of the crystal when it is perfectly aligned to the beam.



3. ANALYSIS

We can now compute the efficiency as the ratio of successfully channeled particles to the total number of incident particles.

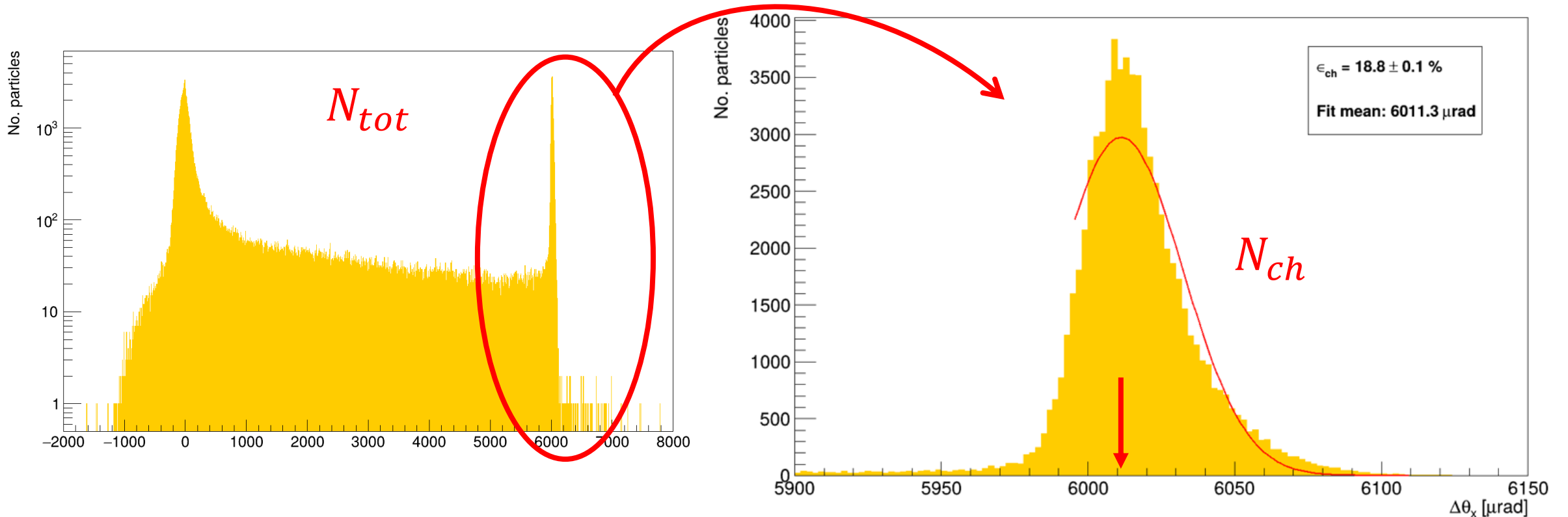


**CHANNELING
EFFICIENCY**

$$\epsilon_{ch} = \frac{N_{ch}}{N_{tot}}$$

3. ANALYSIS

- A Gaussian fit is performed on the deflection peak.
- We fit the right side of the peak to ensure the cleanest measurement (no dechanneled particles contribution).



4. RESULTS

	run	τ_y [$\mu\text{rad}/\text{mm}$]	θ_b [μrad]	ϵ_{ch} [%]
Long crystal 1	8430	3.45	6011	(18.7 ± 0.01)
	8431	3.33	6011	(18.3 ± 0.01)
	8650	3.72	6077	(17.1 ± 0.01)
Long crystal 2	8655	2.97	6124	(17.9 ± 0.01)
	8656	2.83	6123	(17.8 ± 0.01)

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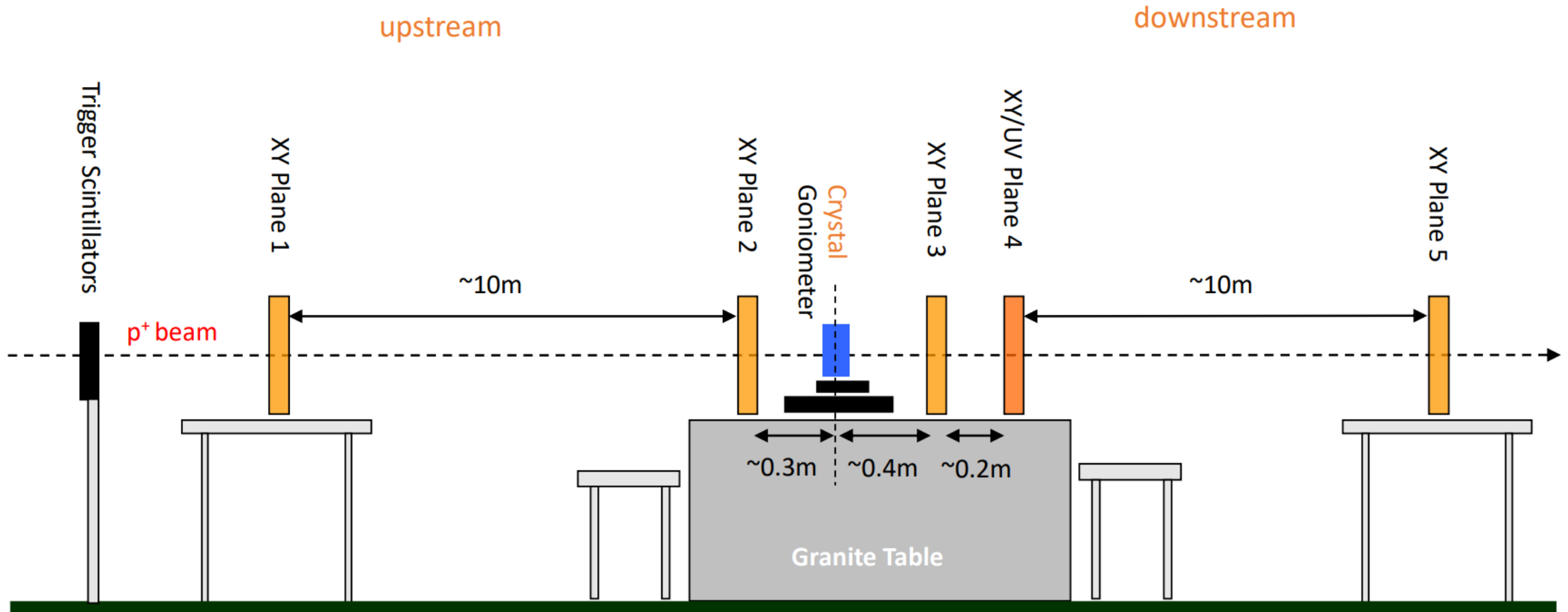
Consistent across different runs for the same crystal

4. RESULTS

	run	τ_y	channeling angle	ϵ_{ch}
Long crystal 1	Run 8430	3.50 $\mu\text{rad}/\text{mm}$	6033 μrad	(18.1 $\pm$ 0.01)%
	Run 8431			
	Run 8650			
Long crystal 2	Run 8655	2.90 $\mu\text{rad}/\text{mm}$	6124 μrad	(17.9 $\pm$ 0.01)%
	Run 8656			

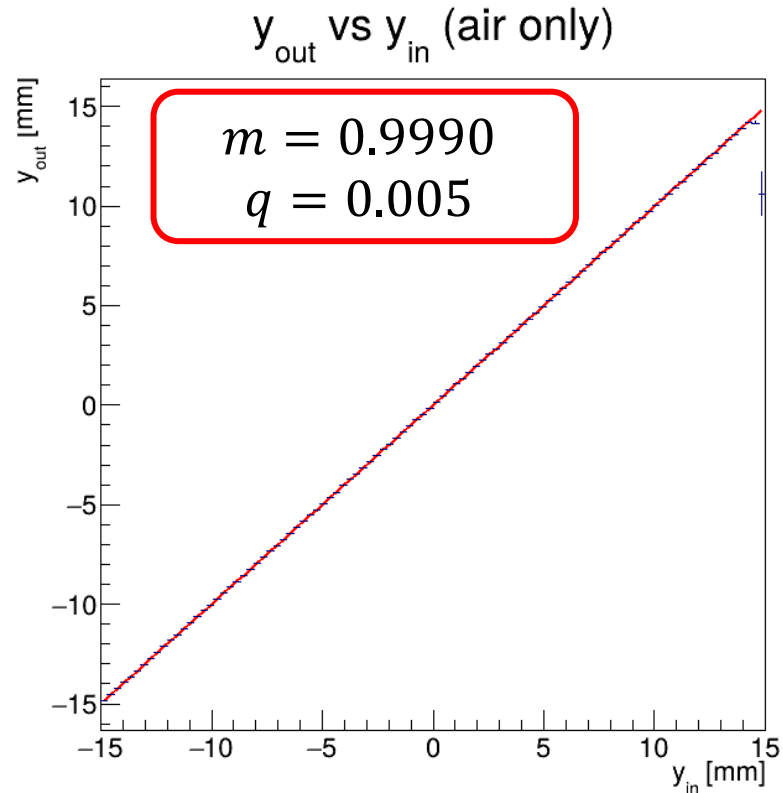
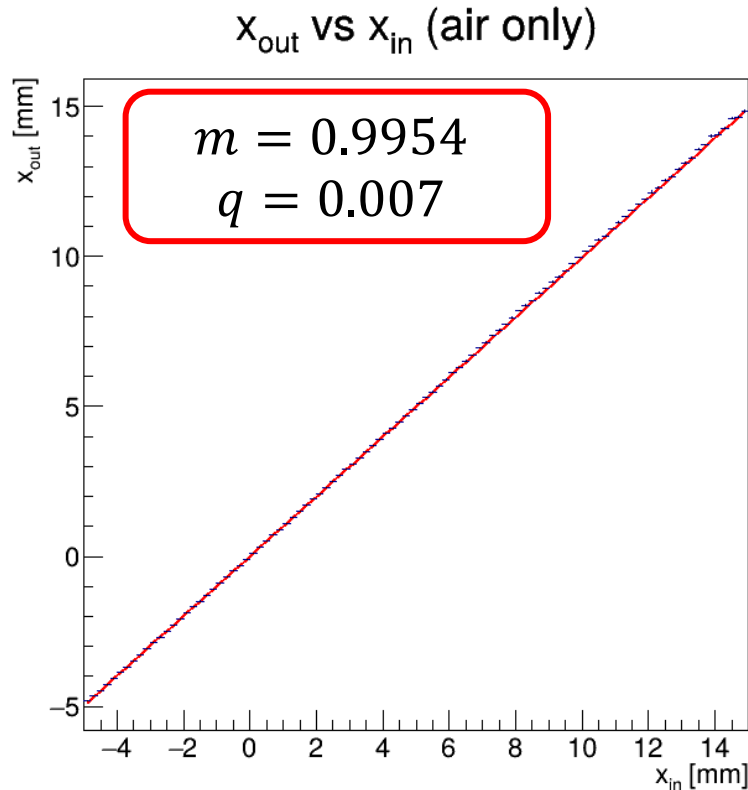
BACKUP SLIDES

BACKUP SLIDES: SCHEMATIC LAYOUT 2

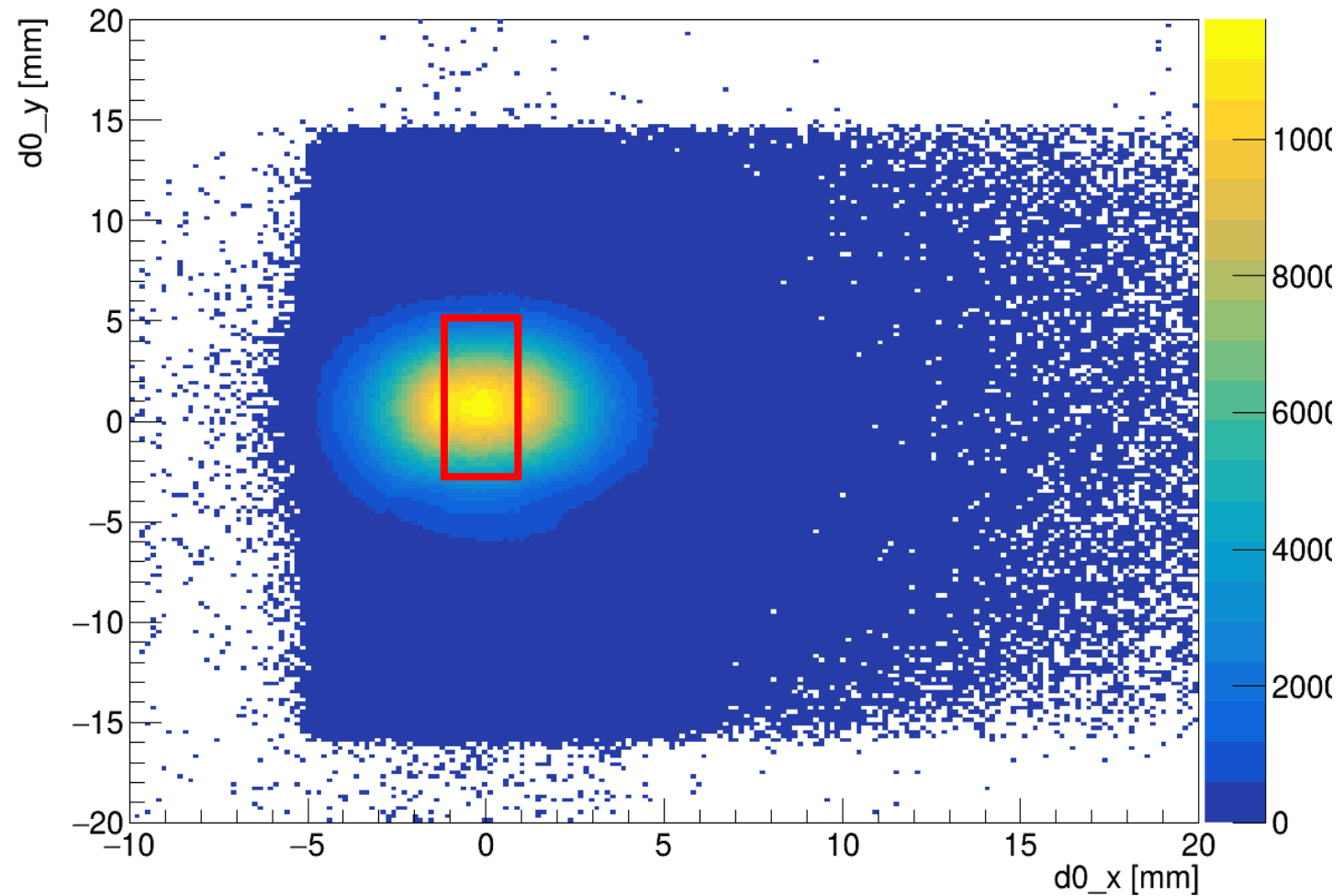


BACKUP SLIDES: ALIGNMENT

We can also check the alignment by doing a linear fit between the upstream and downstream profiles.



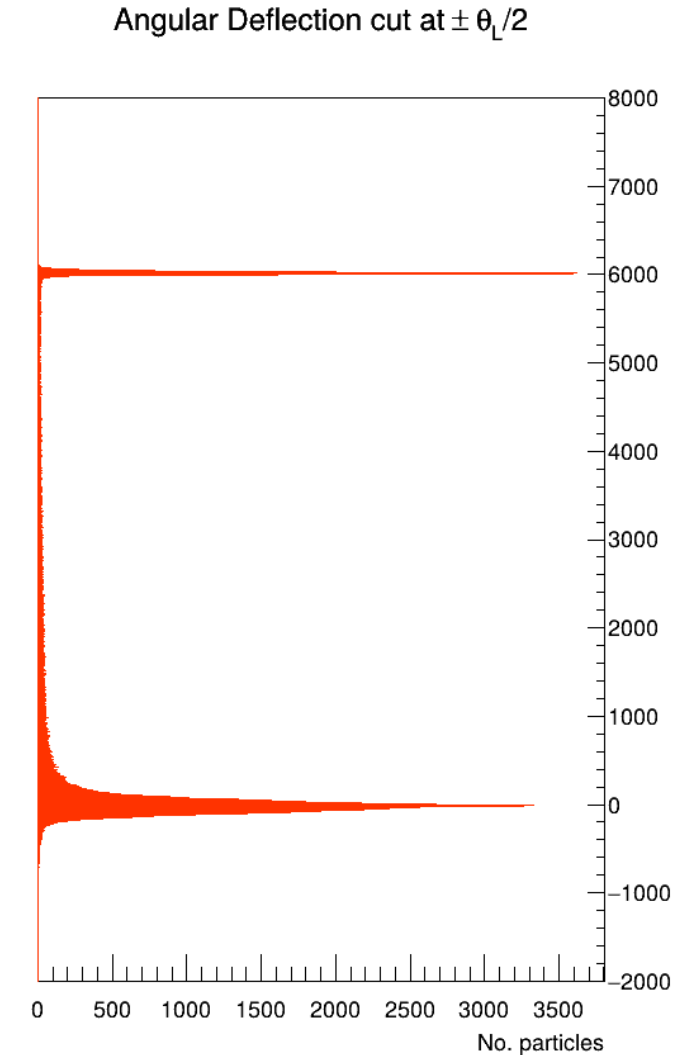
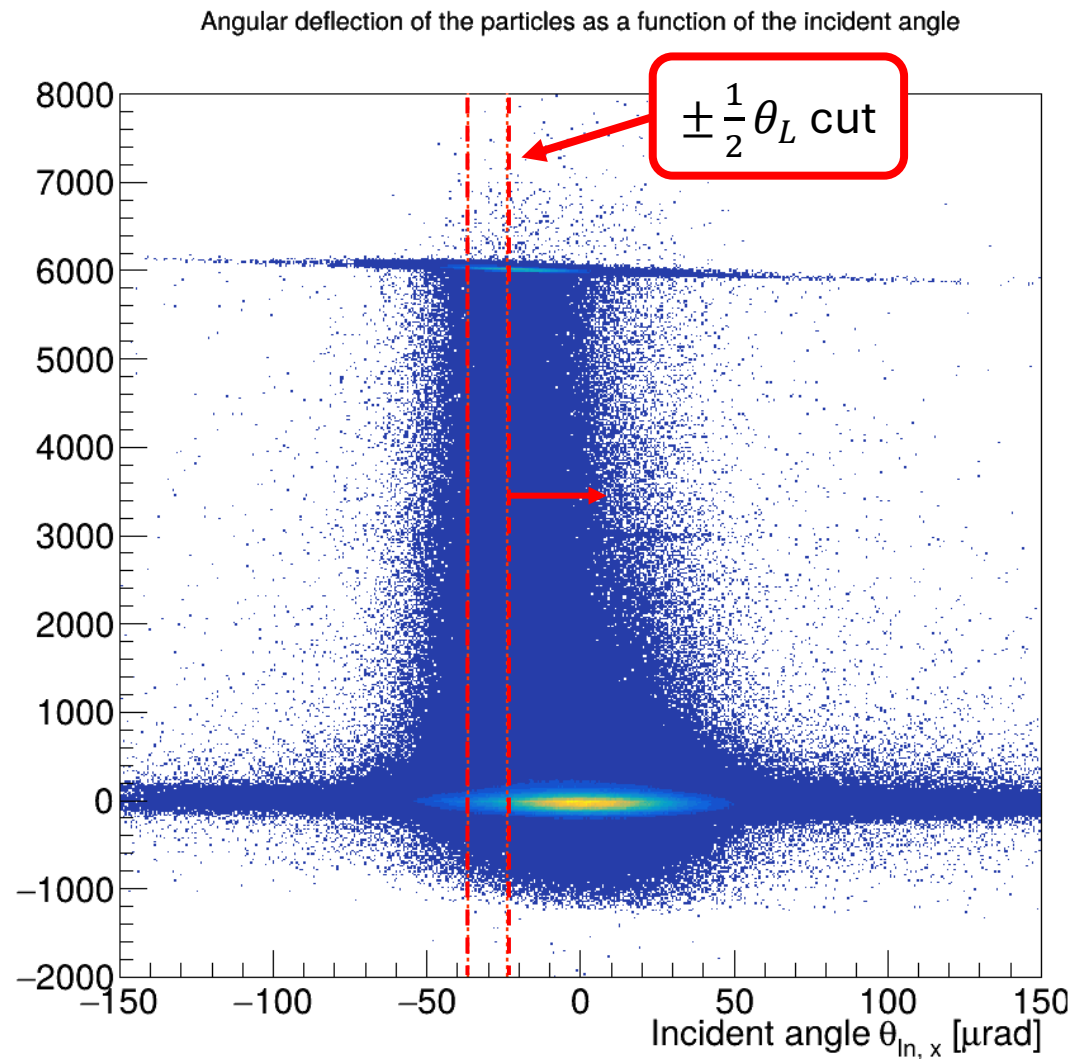
BACKUP SLIDES: SPATIAL CUT



BACKUP SLIDES: ANGLE SELECTION

For each square bin:

- variation of θ_0
- select particles
 $|\theta_{in} - \theta_0| \leq \frac{1}{2} \theta_L$
- compute ϵ_{ch}
- select θ_0 that
maximizes ϵ_{ch}
(through gaussian
fit)

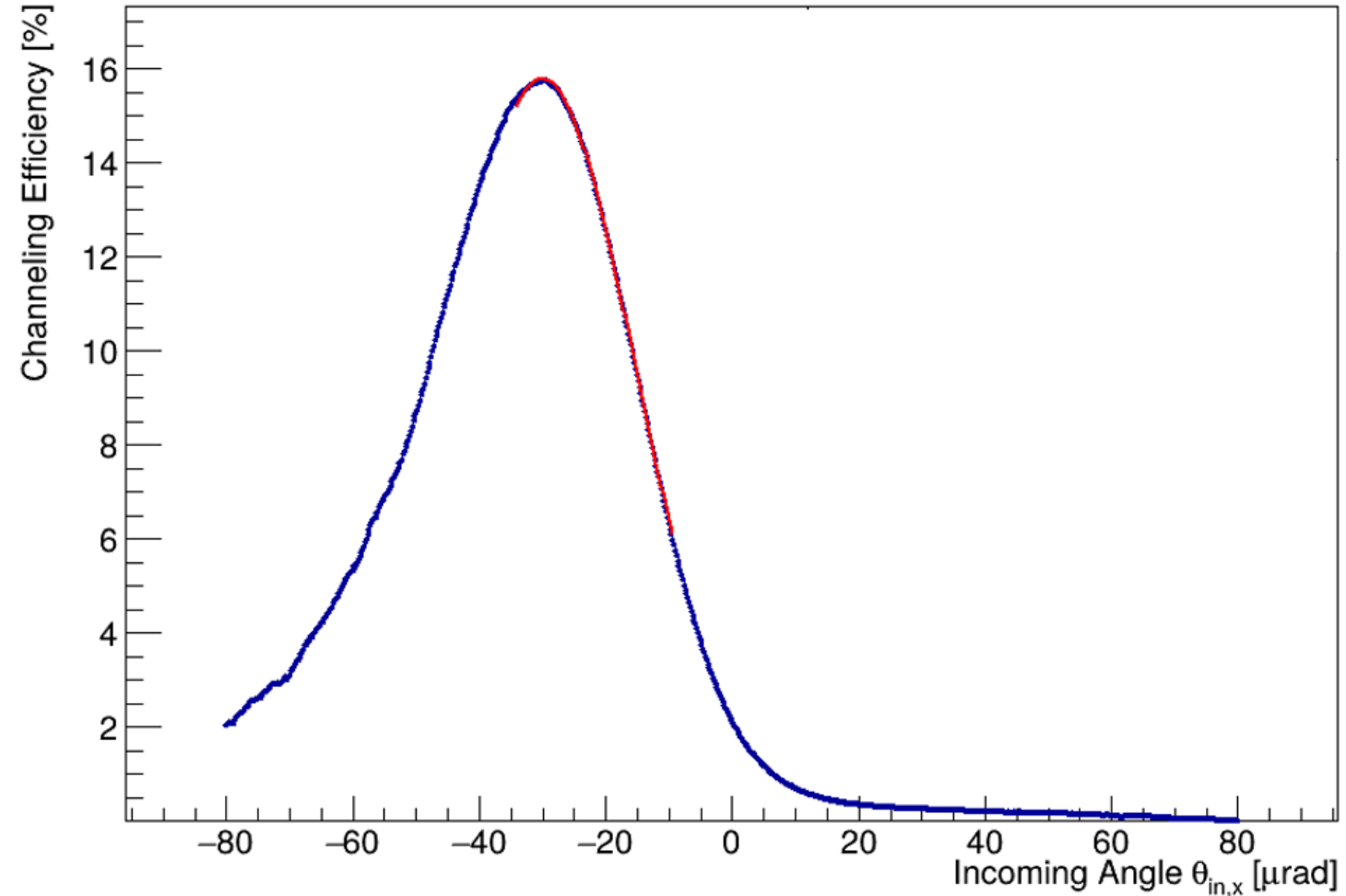


BACKUP SLIDES: ANGLE SELECTION

Angular Acceptance Curve

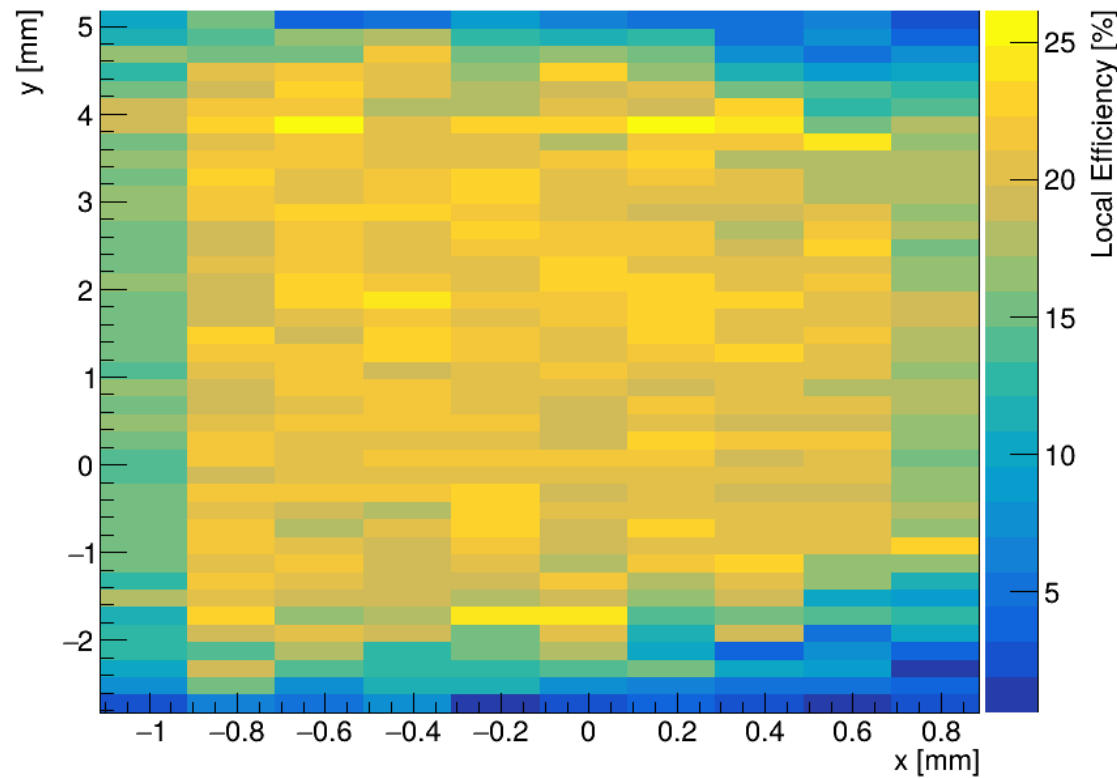
For each square bin:

- variation of θ_0
- select particles
 $|\theta_{in} - \theta_0| \leq \frac{1}{2} \theta_L$
- compute ϵ_{ch}
- select θ_0 that
maximizes ϵ_{ch}
(through gaussian
fit)

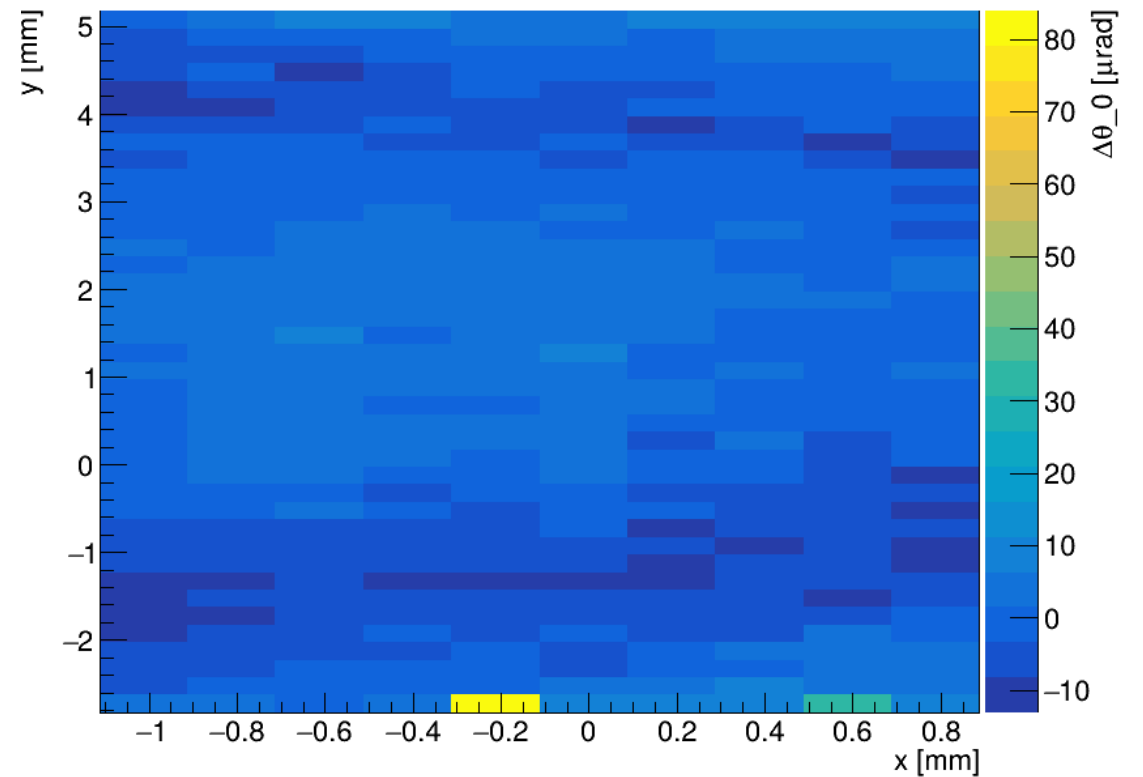


BACKUP SLIDES: TORSION FIT

2D Efficiency Map



Torsion Residuals (Data - parabolic_y)



BACKUP SLIDES: EFFICIENCY CURVE

Crystal efficiency as a function of the incoming angle, after application of the torsion correction. The value of the peak is the efficiency of the crystal when it is perfectly aligned to the beam.

